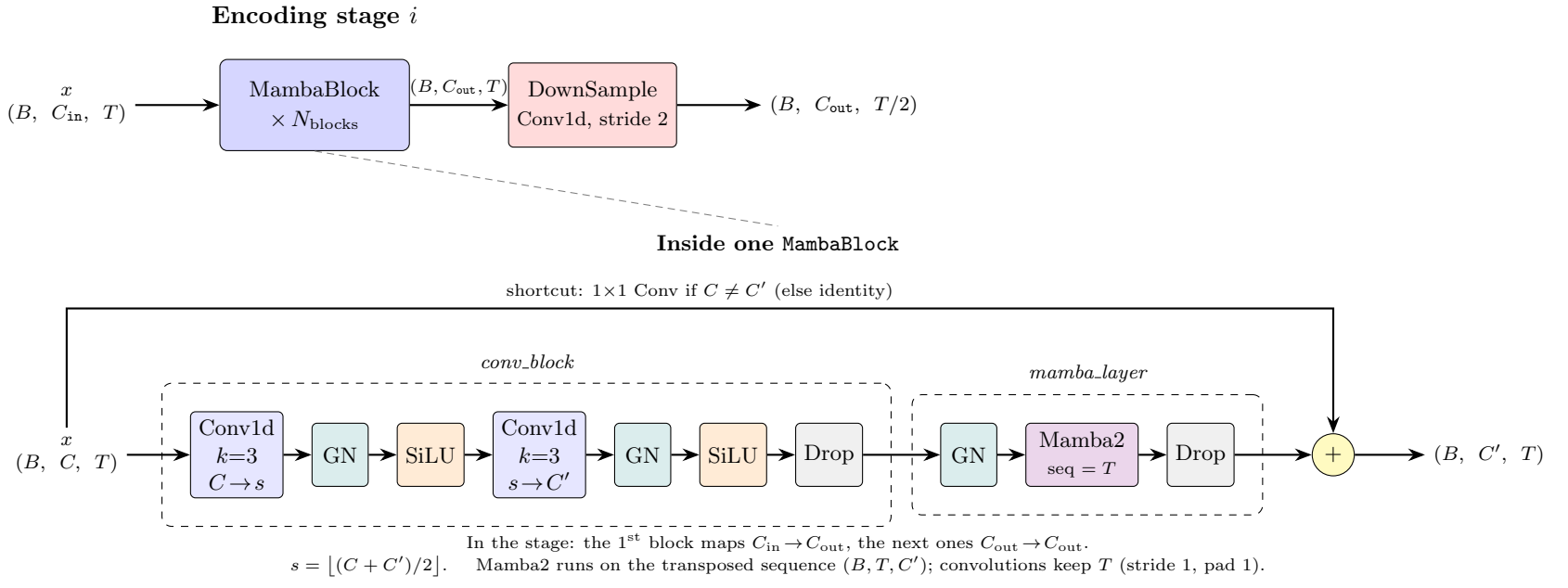

Encoding stage i
x
$(B,\ C_{\text{in}},\ T)$
MambaBlock
$\times N_{\text{blocks}}$
(B, C_{out}, T)
DownSample
Conv1d, stride 2
$(B,\ C_{\text{out}},\ T/2)$
Inside one MambaBlock
shortcut: 1×1 Conv if $C \neq C'$ (else identity)
conv_block
mamba_layer
x
$(B,\ C,\ T)$
Conv1d
k=3
$C \to s$
GN
SiLU
Conv1d
k=3
$s \to C'$
GN
SiLU
Drop
GN
Mamba2
seq = T
Drop
+
$(B,\ C',\ T)$
In the stage: the 1st block maps $C_{\text{in}} \to C_{\text{out}}$, the next ones $C_{\text{out}} \to C_{\text{out}}$.
$s = \lfloor (C + C')/2 \rfloor$. Mamba2 runs on the transposed sequence (B, T, C'); convolutions keep T (stride 1, pad 1).